

مكمل الفيرست

biology 102

chapter 18

Regulation of Gene Expression

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concept 18.1

1) Which of the following molecules is a protein produced by a regulatory gene?

A) operon

- B) inducer
- C) promoter
- D) repressor

Answer: D

2) Which of the following molecules helps to "turn off" genes in a cell?

- A) operon
- B) inducer
- C) promoter
- D) corepressor

Answer: D

3) When taken up by a cell, which of the following molecules binds to a repressor so that the repressor no longer binds to the operator?

- A) inducer
- B) promoter
- C) repressor
- D) corepressor

Answer: A

4) Most repressor proteins are allosteric. Which of the following molecules binds with the repressor to alter its conformation and therefore affect its function?

- A) inducer
- B) promoter
- C) transcription factor
- D) cAMP

Answer: A

5) Which of the following processes would result from a mutation that deactivates a regulatory gene of a repressible operon in an *E. coli* cell?

- A) continuous transcription of the structural gene controlled by that regulator
- B) complete inhibition of transcription of the structural gene controlled by that regulator
- C) irreversible binding of the repressor to the operator

D) continuous translation of the mRNA because of alteration of its structure
Answer: A

6) Which of the following conditions is most likely to cause the lactose operon to be transcribed?

- A) There is more glucose in the cell than lactose.
- B) There is glucose but no lactose in the cell.
- C) The cyclic AMP and lactose levels are both high within the cell.
- D) The cAMP level is high and the lactose level is low.

Answer: C

7) How does the transcription of structural genes in an inducible operon occur?

- A) It occurs continuously in the cell.
- B) It starts when the pathway's substrate is present.
- C) It starts when the pathway's product is present.
- D) It stops when the pathway's product is present.

Answer: B

8) For a repressible operon to be transcribed, which of the following conditions must occur?

- A) A corepressor must be present.
- B) RNA polymerase and the active repressor must be present.
- C) RNA polymerase must bind to the promoter, and the repressor must be inactive.
- D) RNA polymerase must not occupy the promoter, and the repressor must be inactive.

Answer: C

9) Altering patterns of gene expression in prokaryotes would most likely serve an organism's survival by _____.

- A) organizing gene expression, so that genes are expressed in a given order
- B) allowing each gene to be expressed an equal number of times
- C) allowing an organism to adjust to changes in environmental conditions
- D) allowing environmental changes to alter a prokaryote's genome

Answer: C

10) In positive control of several sugar metabolism-related operons, the

cAMP receptor protein

(CRP) binds to DNA to stimulate transcription. Which of the following environmental conditions causes an increase in CRP activity in stimulating transcription?

- A) an increase in glucose and an increase in cAMP
- B) a decrease in glucose and an increase in cAMP
- C) an increase in glucose and a decrease in cAMP
- D) a decrease in glucose and a decrease in the repressor

Answer: B

11) There is a mutation that is found in *E. coli* in the repressor that results in a molecule known

as a super-repressor because it represses the *lac* operon permanently. Which of the following

characteristics would you expect to observe in such a mutant?

- A) It cannot bind to the operator.
- B) It cannot make a functional repressor.
- C) It cannot bind to the inducer.
- D) It makes a repressor that binds CAP.

Answer: C

12) Suppose an experimenter becomes proficient with a technique that allows her to move DNA

sequences within a prokaryotic genome. If a researcher moves the promoter for the *lac* operon to

the region between the beta galactosidase (*lacZ*) gene and the permease (*lacY*) gene, which of the

following results would be most likely?

- A) The three genes of the *lac* operon will be expressed normally.
- B) RNA polymerase will no longer transcribe permease.
- C) The operon will still transcribe the *lacZ* and *lacY* genes, but the mRNA will not be translated.
- D) Beta galactosidase will not be produced.

Answer: D

13) Suppose an experimenter becomes proficient with a technique that allows her to move DNA

sequences within a prokaryotic genome. If a researcher moves the operator

to the far end of the operon, past the transacetylase (*lacA*) gene, which of the following processes would likely occur when the cell is exposed to lactose?

- A) The inducer will no longer bind to the repressor.
- B) The repressor will no longer bind to the operator.
- C) The operon will never be transcribed.
- D) The genes of the *lac* operon will be transcribed continuously.

Answer: D

14) Suppose an experimenter becomes proficient with a technique that allows her to move DNA sequences within a prokaryotic genome. If a researcher moves the repressor gene (*lacI*), along with its promoter, to a position at some several thousand base pairs away from its normal position, which of the following results would be expected?

- A) The repressor will no longer bind to the operator.
- B) The repressor will no longer bind to the inducer.
- C) The *lac* operon will be expressed continuously.
- D) The *lac* operon will function normally.

Answer: D

15) Which of the following results would occur if the repressor of an inducible operon were mutated so that it could not bind the operator?

- A) the irreversible binding of the repressor to the promoter
- B) the reduced transcription of the operon's genes
- C) the continuous transcription of the operon's genes
- D) the overproduction of cAMP receptor protein (CRP)

Answer: C

16) According to the *lac* operon model proposed by Jacob and Monod, what is predicted to occur if the operator is removed from the operon?

- A) The *lac* operon would be transcribed continuously.
- B) Only *lacZ* would be transcribed.
- C) Only *lacY* would be transcribed.

D) Galactosidase permease would be produced, but would be incapable of transporting lactose.

Answer: A

17) Under what conditions does the *trp* repressor block transcription of the *trp* operon?

- A) when the repressor binds to the inducer
- B) when the repressor binds to tryptophan
- C) when the repressor is not bound to tryptophan
- D) when the repressor is not bound to the operator

Answer: B

18) How does extracellular glucose inhibit transcription of the *lac* operon?

- A) by strengthening the binding of the repressor to the operator
- B) by weakening the binding of the repressor to the operator
- C) by inhibiting RNA polymerase from opening the strands of DNA to initiate transcription
- D) by reducing the levels of intracellular cAMP

Answer: D

19) The cAMP receptor protein (CRP) is said to be responsible for positive regulation of the *lac* operon because _____.

- A) CRP binds cAMP
- B) CRP binds to the CAP-binding site
- C) CRP prevents binding of the repressor to the operator
- D) CRP bound to the CRP-binding site stimulates the transcription of the *lac* operon

Answer: D

concept 18.2

20) Imagine that you've isolated a yeast mutant that contains histones resistant to acetylation.

What phenotype would you predict for this mutant?

- A) The mutant will grow rapidly.
- B) The mutant will require galactose for growth.
- C) The mutant will show decreased levels of gene expression.
- D) The mutant will show increased levels of gene expression.

Answer: C

21) Which of the following statements correctly describes the primary difference between enhancers and proximal control elements?

A) Enhancers are transcription factors; proximal control elements are DNA sequences.

B) Enhancers improve transcription; proximal control elements inhibit transcription.

C) Enhancers are located considerable distances from the promoter; proximal control elements are close to the promoter.

D) Enhancers are DNA sequences; proximal control elements are proteins.

Answer: C

22) The reason for differences in the sets of proteins expressed in a nerve and a pancreatic cell of the same individual is that nerve and pancreatic cells contain different _____.

A) genes

B) regulatory sequences

C) sets of regulatory proteins

D) promoters

Answer: C

23) Gene expression is often assayed by measuring the level of mRNA produced from a gene.

What level of the control of gene expression can be analyzed by this type of assay?

A) replication control

B) transcriptional control

C) alternative splicing

D) translational control

Answer: B

24) Which of the following processes would allow the detection of alternative splicing of transcripts from a given gene?

- A) Compare the DNA sequence of the given gene to that of a similar gene in a related organism.
 - B) Measure the relative rates of transcription of the given gene compared to that of a gene known to be constitutively spliced.
 - C) Compare the sequences of different primary transcripts made from the given gene.
 - D) Compare the sequences of different mRNAs made from the given gene.
- Answer: D

25) Which of the following mechanisms is used to coordinate the expression of multiple, related genes in eukaryotic cells?

- A) Environmental signals enter the cell and bind directly to promoters.
 - B) A given gene may have multiple enhancers, but each enhancer is generally associated with only that gene and no other.
 - C) The genes are organized into a large operon, allowing them to be coordinately controlled as a single unit.
 - D) A single repressor is able to turn off several related genes.
- Answer: B

26) DNA methylation and histone acetylation are examples of which of the following processes?

- A) genetic mutation
 - B) chromosomal rearrangements
 - C) epigenetic phenomena
 - D) translocation
- Answer: C

27) Which of the following functions are characteristic of general transcription factors in eukaryotes?

- A) They bind to other proteins or to the TATA box.
- B) They inhibit RNA polymerase binding to the promoter and begin transcribing.
- C) They usually lead to a high level of transcription even without additional

specific

transcription factors.

D) They bind to sequences just after the start site of transcription.

Answer: A

28) How do steroid hormones produce their effects in cells?

A) by binding to the regulatory gene in an operon

B) by activating translation of certain mRNAs

C) by promoting the degradation of specific mRNAs

D) by binding to intracellular receptors and promoting transcription of specific genes

Answer: D

29) How are bacteria able to change their patterns of protein synthesis so quickly in response to environmental changes?

A) mRNAs that are produced are short-lived and degraded within a few minutes of being synthesized.

B) mRNA have long lifespans, allowing the bacteria to use them many times for translation.

C) mRNA is stored for later use.

D) Operons are activated in the presence of transcription factors.

Answer: A

30) A researcher found a method she could use to manipulate and quantify phosphorylation and methylation in embryonic cells in culture. In one set of experiments, she succeeded in increasing acetylation of histone tails in the chromatin of the cells. Which of the following results would she most likely see in these cells?

A) increased chromatin condensation

B) decreased chromatin condensation

C) decreased binding of transcription factors

D) inactivation of the selected genes

Answer: B

31) A researcher found a method she could use to manipulate and quantify phosphorylation and methylation in embryonic cells in culture. One of her colleagues suggested she try increased methylation of cytosine (C) nucleotides in the DNA of promoters of a mammalian system.

Which of the following results would she most likely see?

- A) decreased chromatin condensation
- B) activation of histone tails for enzymatic function
- C) higher levels of transcription of certain genes
- D) inactivation of the selected genes

Answer: D

concept 18.3

32) Which of the following methods is utilized by eukaryotes to control their gene expression that is different from the type of control found in bacteria?

- A) control of chromatin remodeling
- B) control of RNA splicing
- C) transcriptional control
- D) control of both RNA splicing and chromatin remodeling

Answer: D

33) Which of the following processes destroys RNA molecules in a cell if they have a sequence complementary to an introduced double-stranded RNA?

- A) RNA interference
- B) RNA obstruction
- C) RNA blocking
- D) RNA disposal

Answer: A

34) At the beginning of this century, there was a general announcement regarding the sequencing of the human genome and the genomes of many other multicellular eukaryotes. Many people were surprised that the number of protein-coding sequences was much

smaller than they had expected. Which of the following types of DNA make up the rest of the human genome?

- A) DNA that consists of histone coding sequences
- B) DNA that is translated directly without being transcribed
- C) non-protein-coding DNA that is transcribed into several kinds of small RNAs with biological function
- D) non-protein-coding DNA that serves as binding sites for reverse transcriptase

Answer: C

35) Among the newly discovered small noncoding RNAs, one type helps to reestablish methylation patterns during gamete formation and blocks expression of some transposons. Which of the following types of RNA is responsible for this function?

- A) miRNA
- B) piRNA
- C) lncRNA
- D) siRNA

Answer: B

36) Which of the following statements best describes the characteristics of siRNA?

- A) a double-stranded RNA, one of whose strands can complement and inactivate a sequence of mRNA
- B) a single-stranded RNA that can, where it has internal complementary base pairs, fold into cloverleaf patterns
- C) a double-stranded RNA that is formed by cleavage of hairpin loops in a larger precursor
- D) a portion of rRNA that allows it to bind to several ribosomal proteins in forming large or small subunits

Answer: A

37) A researcher introduces double-stranded RNA into a culture of mammalian cells and can identify its location or that of its smaller subsections experimentally, using a fluorescent probe.

Some time later, she finds that the introduced strand separates into single-stranded RNAs, one of

which is degraded. What does this enable the remaining strand to do?

- A) attach to histones in the chromatin
- B) bind to complementary regions of target mRNAs
- C) activate other siRNAs in the cell
- D) bind to noncomplementary RNA sequences

Answer: B

38) A researcher introduces double-stranded RNA into a culture of mammalian cells and can identify its location or that of its smaller subsections experimentally, using a fluorescent probe.

When she finds that the introduced strand separates into single-stranded RNAs, what other

evidence of this single-stranded RNA piece's activity would she find?

- A) She can measure the degradation rate of the remaining single strand.
- B) The rate of accumulation of the polypeptide encoded by the target mRNA is reduced.
- C) The amount of miRNA is multiplied by its replication.
- D) The cell's translation ability is entirely shut down.

Answer: B

concept 18.4

39) The fact that plants can be cloned from somatic cells demonstrates that _____.

- A) differentiated cells retain all the genes of the zygote
- B) genes are lost during differentiation
- C) the differentiated state is normally very unstable
- D) differentiation does not occur in plants

Answer: A

40) Which of the following molecules are involved in pattern formation

during development?

- A) cytoplasmic determinants
- B) miRNAs
- C) cAMP
- D) transcription factors

Answer: A

41) What essential information does the product of the *bicoid* gene in *Drosophila* provide during development?

- A) orientation of the dorsal-ventral axis
- B) orientation of the left-right axis
- C) segmentation
- D) orientation of the anterior-posterior axis

Answer: D

42) What would be the result of a mutation in a maternal effect gene in a female *Drosophila*?

- A) She will not develop past the early embryonic stage.
- B) All of her offspring will show the mutant phenotype, regardless of their genotype.
- C) Only her male offspring will show the mutant phenotype.
- D) Only her female offspring will show the mutant phenotype.

Answer: B

43) Mutations in which of the following genes lead to transformations in the identity of entire body parts?

- A) segmentation genes
- B) egg-polarity genes
- C) homeotic genes
- D) inducers

Answer: C

44) Which of the following are maternal effect genes that control the orientation of the egg and thus the *Drosophila* embryo?

- A) homeotic genes

- B) segmentation genes
- C) egg-polarity genes
- D) morphogens

Answer: C

45) The *bicoid* gene product is normally localized to the anterior end of the embryo. If large amounts of the product were injected into the posterior end as well, which of the following developmental events would occur?

- A) The embryo would grow extra wings and legs.
- B) The embryo would probably show no anterior development and die.
- C) Anterior structures would form in both ends of the embryo.
- D) The embryo would develop normally.

Answer: C

46) Which of the following statements describes a cell that is considered to be differentiated?

- A) The cell replicates by the process of mitosis.
- B) The cell loses connections to the surrounding cells.
- C) The cell produces proteins specific to a particular cell type.
- D) The cell appears to be different from the surrounding cells.

Answer: C

47) When the Bicoid protein is expressed in *Drosophila*, divisions between cells in the embryo are not yet fully developed. This information helps to explain which observation by Nüsslein-Volhard and Wieschaus?

- A) mRNA from the egg is translated into the Bicoid protein.
- B) Bicoid protein diffuses throughout the embryo in a concentration gradient.
- C) Bicoid protein serves as a transcription regulator.
- D) Bicoid protein determines the dorso-ventral axis of the embryo.

Answer: B

48) Which axis in the embryo does the protein product of the *bicoid* gene in *Drosophila*

determine?

- A) anterior-posterior axis
- B) anterior-lateral axis
- C) posterior-dorsal axis
- D) posterior-ventral axis

Answer: A

concept 18.5

49) In colorectal cancer, several genes must be mutated for a cell to develop into a cancer cell.

Which of the following kinds of genes would you expect to be mutated?

- A) genes coding for enzymes that act in the colon
- B) genes involved in control of the cell cycle
- C) genes that are especially susceptible to mutation
- D) genes of the bacteria, which are abundant in the colon

Answer: B

50) Which of the following types of mutation would convert a proto-oncogene into an oncogene?

- A) a mutation that blocks transcription of the proto-oncogene
- B) a mutation that creates an unstable proto-oncogene mRNA
- C) a mutation that greatly increases the amount of the proto-oncogene protein
- D) a deletion of most of the proto-oncogene coding sequence

Answer: C

51) Which of the following processes do normal proto-oncogenes typically exhibit?

- A) They normally suppress tumor growth.
- B) They enhance tumor growth.
- C) They stimulate normal cell growth and division.
- D) They are underexpressed in cancer cells.

Answer: C

52) Which of the following functions does the product of the *p53* gene carry out?

- A) It inhibits the cell cycle.
- B) It slows down the rate of DNA replication by interfering with the

binding of DNA
polymerase.

- C) It causes cells to reduce expression of genes involved in DNA repair.
- D) It allows cells to pass on mutations due to DNA damage.

Answer: A

53) Which of the following statements correctly describes a characteristic of tumor-suppressor gene?

- A) They are frequently overexpressed in cancerous cells.
- B) They are cancer-causing genes introduced into cells by viruses.
- C) They encode proteins that help prevent uncontrolled cell growth.
- D) They often encode proteins that stimulate the cell cycle.

Answer: C

54) Why are *BRCA1* and *BRCA2* considered to be tumor-suppressor genes?

- A) Their normal products participate in repair of DNA damage.
- B) The mutant forms of either one of these prevent breast cancer.
- C) The normal genes make estrogen receptors.
- D) They block penetration of breast cells by chemical carcinogens.

Answer: A

55) Forms of the Ras protein found in tumors usually cause which of the following events to occur?

- A) DNA replication to stop
- B) cell-to-cell adhesion to be nonfunctional
- C) cell division to cease
- D) excessive cell division

Answer: D

56) A genetic test to detect predisposition to cancer would likely examine the *APC* gene for involvement in which type(s) of cancer?

- A) colorectal only
- B) lung and breast
- C) lung only
- D) lung and prostate

Answer: A